Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
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Prof. Francesco Bripi

Risk Management in Financial Institutions (Chapter 23)


Chapter Preview

- ✓ Uncertainty is part of life
- ✓ Entrepreneurs take on risks by creating new products or services
- ✓ Financial Intermediaries (FI) are specialized in understanding and managing risk …
    - … but managing risk is *not easy*
- ✓ Today & next class we will see how FI manage two particular types of risks:
    - credit risk
    - interest rate risk

How Many Types of Risks for FI?

FI face many different types of risks:

- *Credit risk*
- *Interest-rate risk*

This lecture

- *Liquidity risk (large liquidity demand can inflict losses)*
- *Market risk (arises from market price fluctuations)*
- *Foreign exchange risk (can affect value of assets and liabilities)*
- *Sovereign risk (embargoes, defaults, nationalizations)*
- *Technology risk (is the risk that technological investments don't produce anticipated cost savings)*
- *Operational risk (is the risk of fraud, technological failures,*
- *catastrophes, skimming, phishing, hacking)*
- *Insolvency risk (materializes if sudden losses erode capital)*
- *Off-balance-sheet risk (arises from contingent assets and liabilities)*
- *....*

Risk Management

- ✓ Not all risks can (or should) be hedged:
  - there is no way of getting rid of *every possible risk …*
  - … nor this is necessarily desirable (hedging is *costly*)
  - but FIs must be careful not to be exposed to too much risk
    - else they will be bankrupt!

- ✓ FI hence need to analyze very well
  - how risks interact with each other
  - avoid the most severe ones
  - how severe risks that may threaten the existence of the FI can be hedged

This is the central idea behind **risk management**

- ✓ All FI's have a separate Risk Management department

Main topics

- ✓ Managing **CREDIT RISK**
  - Screening (specialization in lending)
  - Monitoring
  - Collateral
  - Credit rationing
  - Long-term relationships and loan commitments
  - The role of collateral for both adv.sel & moral hazard

- ✓ Managing **INTEREST-RATE RISK**
  - Income gap
  - Duration gap

CREDIT RISK

Managing Credit Risk

- ✓ **Credit risk**
  - it is the risk that promised cash flows from loans and securities are **not** paid in full

- ✓ Credit risk arises for two reasons:
  - A. *individual* risks: loans were made to people who shouldn't have received them (*adverse selection* and *moral hazard*)
  - B. *aggregate*/*macro* risks*:* adverse events (such as general economic downturn, or climate extreme events) caused people to default on their obligations

- ✓ FI can't do much about B [*really?*], but can try to fix A:

1	Screening (bank specialization)	4	Credit rationing
2	Monitoring	5	Long term relationship
3	Collateral	6	Loan commitment

1. Screening

- **Screening**
  - identify high quality borrowers, *ex-ante* (= before) lending
  - deny/give credit to un/reliable borrowers
  - reduce *adverse selection*

- Screening can be achieved using
  - **i. Hard** information: info that can be encoded in numbers
    - firm financial statements
    - credit scores, Credit Register (record of borrowers missed payments (*Central Credit Register* at Bank of Italy): all banks see borrower's history in the system
  - **ii. Soft** information: subjective aspects that can't be processed automatically
    - EX: appearance of the prospective borrower, "gut feeling" of the loan officer etc., personal knowledge ("hard working man"), etc…
  - **iii. Specialization**: in a particular type/industry/region of lending
    - Positive side: FIs can do more/better screening
    - Negative side: banks lose benefits of diversification

2. Monitoring

- ✓ **Monitoring** of outstanding loans (and enforcement of covenants)
  - once a loan is obtained, since the payment is fixed in case of no default, incentive to take on *more risk*
  - monitoring is done *ex-post* (= after lending)
  - <u>prevents</u> borrowers from engaging in more risky actions (*moral hazard*)
  - FI may impose and enforce restrictive covenants in loan contract
    - restrict activities borrowers can do with loan: avoid risky projects
  - civil justice (time and costs to recover an unpaid loan)
    - if quality of civil justice is low need more monitoring

3. Collateral

- ✓ Collateral helps banks to reduce losses in case default occurs. Indeed, collateral can reduce
  - **adv. selection**: inducing *self-selection* of borrowers
    - high risk borrowers may realize that they will lose the collateral with high probability and may refrain from applying for a loan
  - **moral hazard**
    - borrowers have something they might lose

- ✓ Loans with collateral are called “secured loans”
  - Ex: *compensating balances*
    - min. $ to keep in a bank account, typically a non-interest-bearing one
    - a form of collateral that also allows better monitoring

- ✓ But research suggests that lender may screen “too little”!
  - when collateral is “**too valuable**”, banks *lose incentive in screening*
  - in a boom: lax lending standards / in a crisis: banks screen more

4. Credit Rationing

Q: Does charging a higher interest rate on loans always increase the bank's profits?

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A: No because the borrowers willing to pay a high interest rate are the riskiest … exactly those the bank would like to avoid (*adverse selection*): better to do **credit rationing**

- ✓ **Credit rationing** means
  - either *refusing total amount* of loan requested by customer
  - or *restricting the size* of the loan requested by customer
  - in both cases: even if customer are willing to pay a high interest rate

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- or *restricting the size* of the loan requested by customer
- in both cases: even if customer are willing to pay a high interest rate

✓ It may be optimal for the bank to ration credit …

- the bank knows that by changing the interest rate it changes the average quality of the pool of borrowers asking for credit
  - if $i \uparrow \Rightarrow$ select only riskier borrowers!!!
  - if $i \downarrow \Rightarrow$ keep high-risk borrowers, but also low-risk ones

Exercise: Credit Rationing

An entrepreneur has a project that yields $1M if it succeeds and 0 otherwise. The probability of success is 10%. In order to implement the project, the entrepreneur needs a loan of $200k and is willing to pay an interest rate of 10%. No collateral.

Q: Should the bank lend to this entrepreneur?

Exercise: Credit Rationing

An entrepreneur has a project that yields $1M if it succeeds and 0 otherwise. The probability of success is 10%. In order to implement the project, the entrepreneur needs a loan of $200k and is willing to pay an interest rate of 20%. No collateral.

Q: Should the bank lend to this entrepreneur?

A: No, the bank shouldn't lend to this entrepreneur

Indeed, the expected profit for the bank is:

$$E(\pi_{bank}) = p \times L \times (1 + i) + (1 - p) \times 0 - L$$

$$E(\pi_{bank}) = 0.1 \times \$200k \times 1.1 + 0.9 \times 0 - \$200k = -\$176k$$

Since this is negative, **no** incentive to lend!

Managing Credit Risk

To *alleviate* asymmetric information problems, FI may use tools that reduce costs of screening and/or of monitoring

✓ (5.) build **long term relationships** with their customers

- repeated interactions alleviate Asy Info
- adv. for borrower: can get lower int. rate, wants to *build* and *keep* good reputation
- adv. for lender: already has soft info of borrower

✓ (6.) impose **loan commitments**

- bank commits to provide up to $X at pre-specified interest rate to firm (a *credit line*)
- adv. for borrower: it gets money when it needs it
- adv. for lender: customer agrees to provide info continuously about their state, even when it does not ask for credit

Managing Credit Risk

Does all this <u>eliminate</u> credit risk?

- ✓ Unfortunately, **no**! Credit risk remains a big issue FI's :
  - ○ Big problem for FI's is not so much *individual* default, but the *aggregate* credit risk (<u>comovement</u> in default rates)
    - ▪ ex.: screening generates much information about *individual* borrower, but little information that the bank could use to understand its *aggregate* credit risk (as a function of macroeconomic state)

- ✓ The recent crisis has raised new questions about credit risk:
  - ○ do **high collateral** values at times make FI's "lazy" about their screening?
  - ○ does **more competition** between banks induce banks to pay less attention to credit risk management?
    - ▪ there is culminating evidence that whenever banks compete fiercely for market share, they lend broadly and screen less...

INTEREST RATE RISK

Managing Interest-Rate Risk

- ✓ Financial institutions, banks in particular, specialize in
  - earning a <u>higher rate of return</u> on their assets
  - *relative to*
  - <u>the interest paid</u> on their liabilities

- ✓ Looks like a pretty simple business! Old joke about bankers "3-6-3" rule:
  - borrow at 3%
  - lend at 6%
  - be on the golf course by 3pm

- ✓ In fact it's not that simple: you have to manage many risks, such as credit risk etc… (see first slide)

- ✓ Here we will look at **interest rate risk**

Managing Interest-Rate Risk

What happens to an FI if $interest\ rates$ change?

Recall: $Return = \frac{C+P_{t+1}-P_t}{P_t} = i + g$

- ✓ Changes in **Income**: interest income on variable-interest assets (loans and other securities) and interest paid on liabilities will change: **Income Gap analysis**
- ✓ Changes in **Net Worth**: values of assets and liabilities on the balance sheet will re-adjust, depending on their *duration:* **Duration Gap analysis**
- ✓ **Important distinction: income ≠ change in value!**

  We will use:
  - ***book*** *values* for Income Gap
  - ***market*** *values* for Duration Gap

Income Gap **analysis**

The idea: look at the balance sheet and ..

- ✓ go over the *asset* side, collect all items whose income flow will adjust with the interest rate**:** ***Rate-Sensitive Assets*** (**RSA**)
  - the items left are the rate-**in***sensitive* assets

- ✓ do the same with the *liability* side: ***Rate-Sensitive Liabilities*** (**RSL**)
  - the items left are the rate-**in***sensitive* liabilities

- ✓ The difference between these two positions (RSA and RSL) will be an indicator by how much your income will change (next slide)

Income Gap analysis

$$\boxed{Income_{gap} = RSA - RSL}$$

Important:

- here we only define it for changes in *1-year* *interest rates* (but you can calculate one income gap for each maturity bucket instead)

✓ RSA (RSL) is the number of dollars of assets (liabilities) that will pay (cost) variable interest rate

✓ *RSA* is a measure of the amount of assets that either reprice, or mature, within 1 year

✓ *RSL* the amount of the liabilities that reprice or mature within 1 year

Income Gap analysis

$$\boldsymbol{Income_{gap} = RSA - RSL}$$

- ✓ <u>Income gap can be positive or negative</u>, depending on the exact composition of the FI's balance sheet

- ✓ Change of $\boldsymbol{NII}$ (*Net Interest Income*):

$$\boxed{\Delta \boldsymbol{NII} = \boldsymbol{Income_{gap}} \times \Delta \boldsymbol{i}}$$

- ✓ So, interest rate changes can **negatively or positively** affect net interest margins, depending on the *<u>sign</u>* of $Income_{gap}$
  - if $Income_{gap} > 0$: banks make more money if interest rates rise (but loses out if interest rates fall)
  - the reverse if $Income_{gap} < 0$

JP Morgan Chase, Dec2015

Assets		Liabilities	
RSA (repriceable or that mature <1Y)	$1,238 B	RSL	$426 B
		of which:	
		-deposits<1Y	$284 B
		-long-term debt reprices <1Y	$121 B
		-variable rate preferred stocks	$18.1 B
		-long-term debt matures <1Y	$2.7 B
Non-RSA	$1,113 B	Non-RSL	$1,677 B
		Total Liabilities	$2,103 B
Total Assets	$2,351 B	Equity Capital	$248 B

Exercise: JP Morgan Chase

Q1: what is JPM's income gap?

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A1: $Income_{gap} = \$1{,}238\ B\ - \$426B = \$812\ B$

Q2: if interest rates increase by 50 bps, how does net interest income change for JPM Chase?

Exercise: JP Morgan Chase

Q1: what is JPM's income gap?

A1: $Income_{gap} = \$1{,}238\ B\ - \$426B = \$812\ B$

Q2: if interest rates increase by 50 bps, how does net interest income change for JPM Chase?

A2: $\Delta i = .5\%$, $\Delta NII = \boldsymbol{Income_{gap}} \times \Delta \boldsymbol{i}$

$$\Delta NII = \$812\ B \times 0.005 = \$4{,}06\ B$$

Exercise: JP Morgan Chase

Q1: what is JPM's income gap?

A1: $Income_{gap} = \$1{,}238\ B\ - \$426B = \$812\ B$

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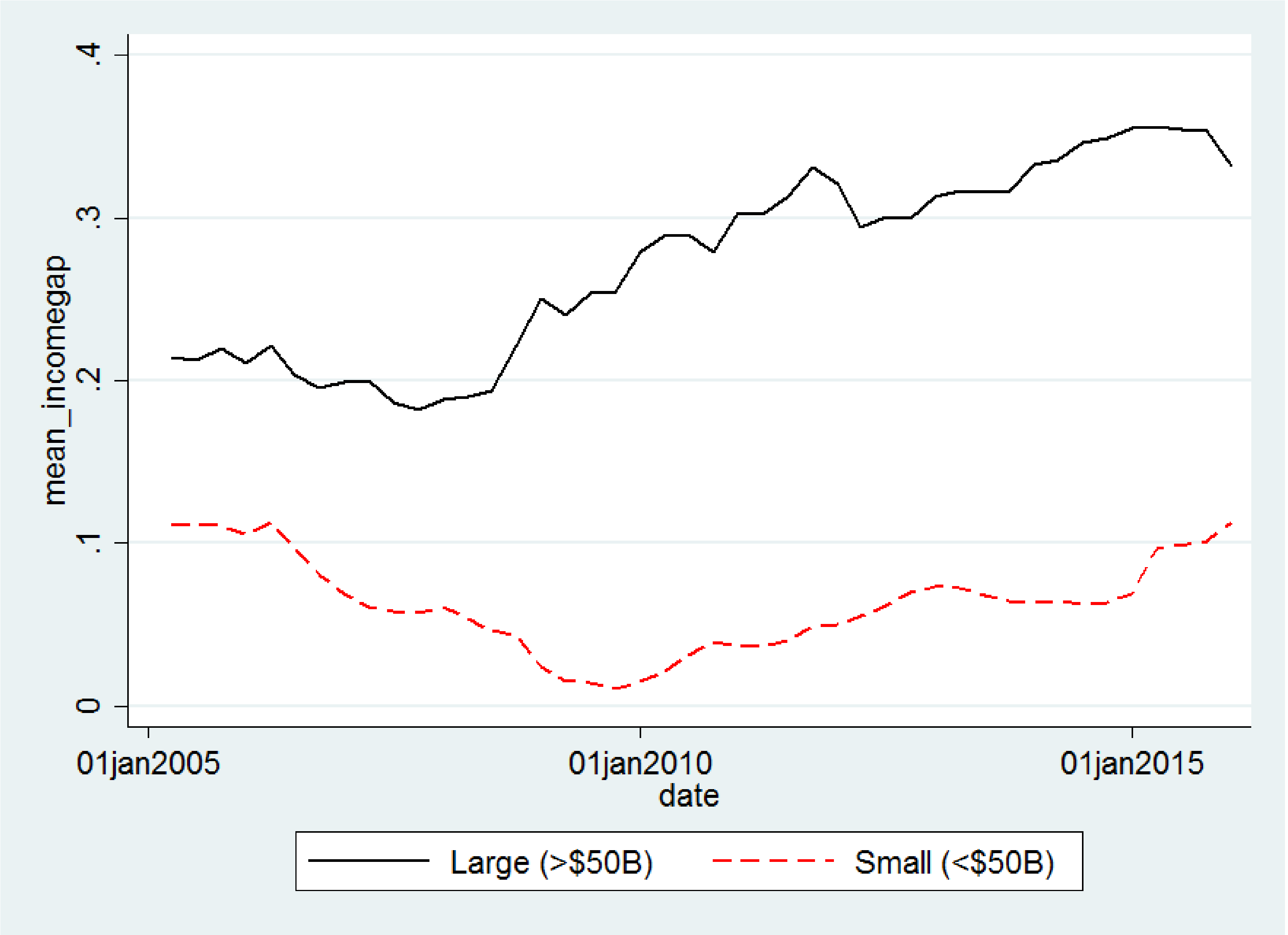
✓ Note that:

- 53% of the asset side of JPM's balance sheet is RSA, but only 20% of liabilities are RSL.
- in this case RSA>RSL, often the reverse: RSA<RSL …
- as i ↑, overall position (of RSA Vs RSL) of the bank matters!

Average Income Gap/Assets, US Banks



Income Gap analysis

- ✓ Those are the actual numbers from US banks balance sheets (named *call reports*)

- ✓ It is generally complicated to determine RSA and RSL

- ✓ Need to find those assets & liabilities whose income will change (because the interest rates will be reset) within the year

- ✓ Some items are obvious (those with maturity <1year for example) but <u>others also may be affected</u> …
  … look at next call report!

First National Bank

Assets		Liabilities	
Reserves and cash items	$5 million	Checkable deposits	$15 million
Securities		Money market	
Less than 1 year	$5 million	deposit accounts	$5 million
1 to 2 years	$5 million	Savings deposits	$15 million
Greater than 2 years	$10 million	CDs	
Residential mortgages		Variable rate	$10 million
Variable rate	$10 million	Less than 1 year	$15 million
Fixed rate (30-year)	$10 million	1 to 2 years	$5 million
Commercial loans		Greater than 2 years	$5 million
Less than 1 year	$15 million	Fed funds	$5 million
1 to 2 years	$10 million	Borrowings	
Greater than 2 years	$25 million	Less than 1 year	$10 million
Physical capital	$5 million	1 to 2 years	$5 million
		Greater than 2 years	$5 million
		Bank capital	$5 million
Total	$100 million	Total	$100 million

Determine the RSA

- ✓ Any asset with a maturity below 1 year will change its <u>interest income</u> if interest rate changes
  - o *securities and commercial loans with maturity < 1Y*

- ✓ Some assets have mat. >1Y but are adjusted automatically
  - o ***Variable rate mortgages*** (borrowers pay x%+LIBOR)

- ✓ But also <u>income</u> on ***30-year fixed rate mortgages***, even if they do not mature within 1 year, can change
  - o Indeed, homeowners can decide:
    - ▪ to *repay early* by selling their homes or *refinance* mortgage at lower rate (if house price increases)
    - ▪ to *move mortgage* to another bank offering new lower rate (if int. rate falls)

- ✓ Income from securities with a term to maturity of > 1 year <u>will not change</u>

Determine the RSL

- ✓ Any liability with a maturity<1Y will change its interest income (*expense* for the bank) if interest rate changes
  - *money market deposits*
  - *variable rate CDs and CDs below 1Y*
  - *Fed Funds*
  - *other short term borrowings*

- ✓ *Checking* and *savings account* have interest rate that can be changed at any time by the bank, although usually banks keep them constant
  - partially RSL

First National Bank

Assets		Liabilities	
Reserves and cash items	$5 million	Checkable deposits	$15 million
Securities		Money market	
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obvious		Greater than 2 years	$5 million
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		Greater than 2 years	$5 million
		Bank capital	$5 million
Total	$100 million	Total	$100 million

obvious Less obvious

Income Gap for First National Bank

EX: determine change in *Net Interest Income* if i increases by 5 pct. points (Δi = 5%=0.005)

- ***RSA*** = \$5M + \$10M + \$15M + 20% × \$10M =\$32M
- ***RSL*** = \$5M + \$25M + \$5M + \$10M + 10% × \$15M + 20% × \$15M = \$49.5M
- $\boldsymbol{Income_{gap}}$ = *RSA* – *RSL* **=** \$32.0M – \$49.5M = –\$17.5M
- $\Delta \boldsymbol{NII} = Income_{gap} \times \Delta i =$ \$1.6M – \$ 2.5M = –\$ 0.9M

✓ Alternatively:

- interest income = +5% × \$32.0M = +\$ 1.6M
- interest expense = +5% × \$49.5M = +\$ 2.5M
- $\Delta \boldsymbol{NII} =$ \$1.6M – \$ 2.5M **= –\$ 0.9M**

If ***RSL > RSA: i*** ↑ results in ***NII*** ↓

Pitfalls of Income Gap Analysis

Problems with income gap analysis:

1. <u>short-term focus</u>: mat.<1year is **coarse**!
   - EX.: a security with 1.5 years residual maturity could also be a RSA!

2. if interest rates rise, the **asset <u>values</u>** of bonds, loans and securities values with <u>long maturities</u> will <u>*fall*</u>!
   - particularly relevant for *mark-to-market* accounting!
   - if securities <u>fall in value</u> the corresponding amount will have to be <u>written off</u>, reducing the bank's net worth

Duration Gap analysis helps with this

it provides an <u>index</u> that captures the
sensitivity of a bank's net worth to interest rate risk

Duration Gap **analysis**

- ✓ The idea:
  1. calculate the duration of the FI’s assets
  2. do the same for its liabilities
  3. then, compare the two to determine the duration of the bank’s net worth

- ✓ Recall duration: $DUR = \sum_{t=1}^{n} t \cdot \frac{PV_t}{PV_{TOT}} = \frac{\sum_{t=1}^{n} t \cdot \frac{CF_t}{(1+i)^t}}{\sum_{t=1}^{n} \frac{CF_t}{(1+i)^t}}$

- ✓ If the change in interest is small, then:

$$\%\Delta P = \frac{\Delta P}{P} \approx -DUR \times \frac{\Delta i}{1+i}$$

- ✓ Idea: similar approach here, but $\%\Delta P$ (rate of capital gain) is replaced with changes in the bank’s net worth

Duration Gap analysis: derivation

✓ Recall that duration is **additive**

- the duration of a portfolio (*P*=*J*+*K*) is just a *(market) value* **weighted average** of duration of each asset:

$$DUR_p = \frac{MV_J}{MV_J + MV_K} \times DUR_J + \frac{MV_K}{MV_J + MV_K} \times DUR_K$$

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$$DUR_p = \frac{MV_J}{MV_J + MV_K} \times DUR_J + \frac{MV_K}{MV_J + MV_K} \times DUR_K$$

  - by definition: $NW = A - L$, so $A = NW + L$
  - hence if you see *assets* as a portfolio of *Net Worth* + *Liabilities:*

$$DUR_A = \frac{NW}{A} \times DUR_{NW} + \frac{L}{A} \times DUR_L$$

Duration Gap analysis: derivation

✓ Recall that duration is **additive**

- the duration of a portfolio (*P*=*J*+*K*) is just a *(market) value* **weighted average** of duration of each asset:

$$DUR_p = \frac{MV_J}{MV_J + MV_K} \times DUR_J + \frac{MV_K}{MV_J + MV_K} \times \ DUR_B$$

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- hence if you see *assets* as a portfolio of *Net Worth* + *Liabilities:*

$$DUR_A = \frac{NW}{A} \times DUR_{NW} + \frac{L}{A} \times DUR_L$$

- it follows that:

$$DUR_{NW} = \frac{A}{NW} \times \left(DUR_A - \frac{L}{A} \times DUR_L \right)$$

Duration Gap analysis: derivation

- ✓ Define

$$\boxed{DUR_{GAP} \equiv DUR_A - \frac{L}{A} \times DUR_L}$$

- ✓ So, we have

$$DUR_{NW} = \frac{A}{NW} \times DUR_{GAP}$$

- ✓ Finally since

$$\frac{\Delta NW}{NW} \approx -DUR_{NW} \times \frac{\Delta i}{1+i} \Rightarrow \frac{\Delta NW}{NW} \approx -\left(\frac{A}{NW} \times DUR_{GAP}\right) \times \frac{\Delta i}{1+i}$$

- ✓ We arrive at

$$\boxed{\frac{\Delta NW}{A} \approx -DUR_{GAP} \times \frac{\Delta i}{1+i}}$$

Formulas you need to remember!!!

Duration Gap Analysis

$$\frac{\Delta NW}{A} \approx -\Big(\overbrace{DUR_A - \frac{L}{A} \times DUR_L}^{= DUR_{GAP}}\Big) \times \frac{\Delta i}{1+i}$$

✓ Definition: DUR_{GAP} measures the *sensitivity* of a bank's Net Worth (relative to the total value of assets) in response to changes in interest rates

✓ *Important assumption*: Δi is the same for all assets & liabilities (regardless of maturity structure)

✓ *Recall*: values are market based!

Duration Gap

- ✓ <u>Duration gap</u> can be <u>positive</u> or <u>negative</u>, depending on the exact composition of the FI's balance sheet
  - so, interest rate changes can be **negatively or positively related** to changes in net worth
  - if $DUR_{GAP} > 0$ (normal case)
    - liabilities have *relatively* <u>shorter</u> duration than assets
    - as $i \uparrow \Longrightarrow NW \downarrow$
    - in order to change this, the FI may sell assets with high duration and convert them into assets with lower duration, even though typically FI's banks earn from mismatch: long duration assets and short duration liabilities!
  - if $DUR_{GAP} < 0$
    - liabilities have *relatively* <u>longer</u> duration than assets
    - as $i \uparrow \Longrightarrow NW \uparrow$
- ✓ In either case, the duration gap measures FI's **net worth** exposure to interest-rate risk

Duration Gap analysis: exercise 1

Consider a market value (!) bank balance sheet

Assets		Liabilities	
Rate-sensitive assets (Duration of 0.5 years)	€50	Rate-sensitive liabilities (Duration of 0.3 years)	€80
Rate-insensitive assets (Duration of 5 years)	€50	Rate-insensitive liabilities (Duration of 5 years)	€10
		Bank capital	€10
Total	€100	Total	€100

Q1: what is this bank's duration gap?

Duration Gap analysis: exercise 1

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(Duration of 0.5 years)		(Duration of 0.3 years)	
Rate-insensitive assets	€50	Rate-insensitive liabilities	€10
(Duration of 5 years)		(Duration of 5 years)	
		Bank capital	€10
Total	€100	Total	€100

Q1: what is this bank's duration gap?

A1: $$DUR_A = 0.5 \times \frac{50}{100} + 5 \times \frac{50}{100} = 2.75$$

$$DUR_L = 0.3 \times \frac{80}{90} + 5 \times \frac{10}{90} = 0.82$$

$$DUR_{GAP} = DUR_A - \frac{L}{A} DUR_L = 2.75 - \frac{90}{100} \times 0.82 = 2.01$$

Duration Gap analysis: exercise 1

Q2: Assume that interest rates go up from 10% to 11%. What will happen to this bank's net worth?

Duration Gap analysis: exercise 1

Q2: Assume that interest rates go up from 10% to 11%. What will happen to this bank's net worth?

A2:

$$\frac{\Delta NW}{A} = -DUR_{GAP} \times \frac{\Delta i}{1+i}$$
$$= -2.012 \times \frac{0.01}{1+0.1} \approx -1.83\%.$$

- with total assets of €100, this translates into a €1.83 negative change in the FI's net worth …
- … almost 20% of the entire net worth of €10! A lot of insolvency risk because the bank has high leverage (90%) to start with

Q3: What if interest rates go down from 10% to 9%?

Duration Gap analysis: exercise 1

Q1: Assume that interest rates go up from 10% to 11%. What will happen to this bank's net worth?

A1:

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- … almost 20% of the entire net worth of €10! A lot of insolvency risk because the bank has high leverage (90%) to start with

Q3: What if interest rates go down from 10% to 9%?

A3: $\Delta NW/A = +€1.83$ (note: same magnitude, opposite sign!)

Targeting Duration Gap

- ✓ Note that if an FI perfectly matched the duration of assets and liabilities then it would have little IRR ($\boldsymbol{DUR_{GAP} \approx 0}$)

- ✓ But this is unlikely:
  - banks earn from the natural mismatch between long duration assets and short duration liabilities!
  - thus, for a bank hedging IRR by altering the balance sheet can be costly (*lose profit margin*)

- ✓ Alternatively, the bank hedges by taking an (opposite) position in the <u>derivatives market</u>
  - interest rate *futures/forwards* (see Lect. 9 and 10)
  - *interest rate swaps* (covered in chap. 24, optional)

Exercise 2: Targeting Duration Gap

Consider the following incomplete balance sheet:

Assets		Liabilities	
Loans	€200	Deposits	€200
(Duration of 5 years)		(Duration of 1 year)	
Securities	€300	Commercial paper	
(Duration of 3 years)		(Duration of 0.5 years)	
Cash	€500	Bank capital	
Total	€1000	Total	€1000

Note that cash has a 0 duration

The manager wants to achieve a duration gap of 1.4 years

Q: How much commercial paper does the bank need to raise? What is this bank's net worth in this case?

Exercise 2: Targeting Duration Gap

Consider the following incomplete balance sheet:

Assets		Liabilities	
Loans	€200	Deposits	€200
(Duration of 5 years)		(Duration of 1 year)	
Securities	€300	Commercial paper	
(Duration of 3 years)		(Duration of 0.5 years)	
Cash	€500	Bank capital	
Total	€1000	Total	€1000

Note that cash has a 0 duration.

The manager wants to achieve a duration gap of 1.4 years

Q: How much commercial paper does the bank need to raise? What is this bank's net worth in this case?

A: denote with *X* the amount of CP:

$$1.4 = \frac{200}{1000} \times 5 + \frac{300}{1000} \times 3 - \frac{200 + X}{1000} \times \left(\frac{200}{200 + X} \times 1 + \frac{X}{200 + X} \times 0.5\right).$$

X=CP

Solve for X:

X=€600

Hence: NW= € 200